

cont. off policy

off policy learning — at least two cases — agent is acting using B trying to learn π
 — someone else uses B well, agent learns π from B
 behavior policy — B
 target policy — π $\rightarrow$ on-policy

Behavior Policy B : a lot of ep. of training
 $ep_i \rightarrow ep_n$ can be large amounts of ep.
 can learn discounted return at any time t
 $G_{t:B} = \text{discounted return}$

Target Policy π : perform actions $\uparrow$ but learn values here
 $G_B(s,a) \Rightarrow G_\pi(s,a)$

Question: Compute $V_B(s)$ and $G_B(s,a)$

From $V_B(s) \xrightarrow{\text{learn}} V_\pi(s)$

From $G_B(s,a) \xrightarrow{\text{learn}} G_\pi(s,a)$
 acting here $\longrightarrow$ learning this

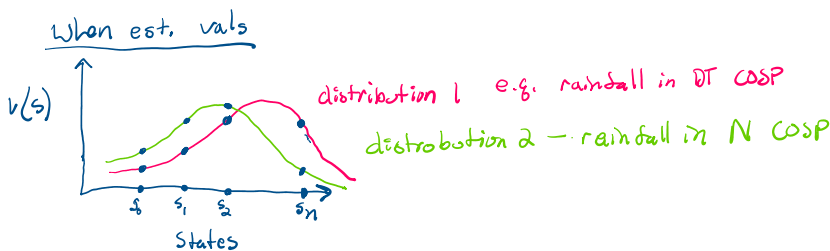
Computing G_t or Return values is necessary for learning $v(s)$ or $q(s,a)$ values.

Agent assumes the ep. is a state-action seq. is exactly the same in B and π

$$G_{t,\pi} = \overset{\text{row}}{P} * G_{t,B}$$

scaling / weighted factor

important sampling factor? Claude find this in the book, the prof was unsure



take original values in one distro and $*$ by a weighted factor to est. another distro

$$v_\pi(s) = \overset{\text{est val}}{\downarrow} \mathbb{E}[G_{t,\pi} | S_{t,\pi} = s]$$

Scale B to get π

$$= \mathbb{E}[P * G_{t,B} | S_t = s]$$

$$S_{t,\pi} = S_{t,B} = s \text{ or } S_t = s$$

row

What is the value of P : the weight or the important sampling factor for transfer from G_t values from B policy to target policy?

$f_t = \text{Probability (of states and actions in target policy } \pi \text{ in the imaginary ep. in time } t \text{ to end time } T)$

$$f_t = \frac{\text{Probability (of states and actions in target policy } \pi \text{ in the imaginary ep. in time } t \text{ to end time } T)}{\text{Prob (states and actions in behave policy } B \text{ in the real ep. from } t \text{ to } T)}$$

$$\begin{aligned} p_t &= \frac{\text{Prob}(a_{t,\pi} | s_{t,\pi}) * \text{Prob}(a_{t+1,\pi} | s_{t+1,\pi}) * \dots}{\text{Prob}(a_{t,B} | s_{t,B}) * \text{Prob}(a_{t+1,B} | s_{t+1,B}) * \dots} \\ &= \frac{\pi(a_t | s_t) * \pi(a_{t+1} | s_{t+1}) * \pi(\dots) * \pi(a_{T-1} | s_{T-1})}{B(a_t | s_t) * B(a_{t+1} | s_{t+1}) * B(\dots) * B(a_{T-1} | s_{T-1})} \end{aligned}$$

$$\prod_{k=t}^{T-1} \frac{\pi(a_k | s_k)}{B(a_k | s_k)}$$

$$\begin{aligned} G_{t,\pi} &= \prod_{k=t}^{T-1} \frac{\pi(a_k | s_k)}{B(a_k | s_k)} G_{t,B} \\ &= \rho_t G_{t,B} \quad \text{importance sampling ratio} \end{aligned}$$

We now have a formula to transfer $G_{t,B}$ to $G_{t,\pi}$
or transfer values from B to π

Next Question: to actually transfer compute $V_\pi(s)$ i.e. values of states in target policy
($Q_\pi(s,a)$ i.e. values of state-action pairs in target Policy)

$$V_\pi(s) = \mathbb{E} \left[\underset{\substack{\uparrow \\ \text{transfer | importance} \\ \text{ratio or weight}}}{\rho_t} * G_{t,B} \mid S_t = s \right]$$

From the book, gives two ways to compute $V_\pi(s)$

- ① ordinary sampling
- ② Weighted sampling

Ordinary Sampling

$$\tau = \tau_0$$

define $\tau(s)$ = the set of time points at which state s occurs
in ep. of training using B policy

prob draws the standard 2x3 maze

then $|\tau(s)|$ = # of times state s occurs in the
ep. of training using behavior policy B

$$V_\pi(s) = \sum_{t \in \tau(s)} \rho_t G_{t,B}$$

$$V_{\pi}(s) = \frac{\sum_{t \in \tau(s)} p_t G_{t,s}}{|\tau(s)|}$$

ex. $\rightarrow$ Spse we want to compute $V_{\pi}(s_2) = [G_t | s_t = s_2] * p$

Weighted Sampling

$$V_{\pi}(s) = \frac{\sum_{t \in \tau(s)} p_t G_{t,s}}{\sum_{t \in \tau(s)} p_t}$$

5.6 Incremental implementation for MC algorithms
 we did not go over this but we should know it
 either presentations next week,